

Owen Li

Mechatronics Engineer

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Experience

Generac

[05/2026 - 08/2026]

Electrical Engineer

- * Designed and built facility ethernet communication network for asset testing controls
- * Designed 800V fuse inverter retrofit to ensure product spec compliance
- * Created inventory forecasting software to assist buyer in inventory management

Form Energy

[07/2023 - 12/2023]

Test Engineer

- * Tested thermal performance of 400A contactors to identify failure modes
- * Led component component selection based on test results
- * Designed and prototyped electrical box containing contactor electrical system
- * Navigated around size and cost constraints allowing for adoption in next generation test systems

Harbinger Motors

[01/2023 - 04/2023]

Electrical Engineer

- * Visualized entire vehicle wire harness layouts using CATIA for 3 vehicle wheelbases
- * Created engineering drawings for wire harness layouts for supplier communication
- * 3D printed 15+ PCB enclosures, test jigs, and sensor mounts
- * Designed position validation BMS analog front end using Altium

Ford Motors

[07/2021 - 12/2021]

Firmware Engineer

- * Created Python unit test script for infotainment system using pytest with email updates of test results
- * Utilized Arduino UNO and relays to simulate vehicle ignition during testing

Solar Car Team

[09/2020 - 04/2024]

Battery System Lead

- * Led the spec, design, manufacture, and testing of 36S8P, 5.2kWh, 131V nominal li-ion pack
- * Implemented modular design consisting of 9 4S8P modules connected in series
- * Conducted IR and thermal testing of individual cells and modules

Skills

Mechanical: SOLIDWORKS, CATIA, Machine Shop, 3D printing

Electrical: Altium, Oscilloscope, Multimeter, Power Supply, Soldering

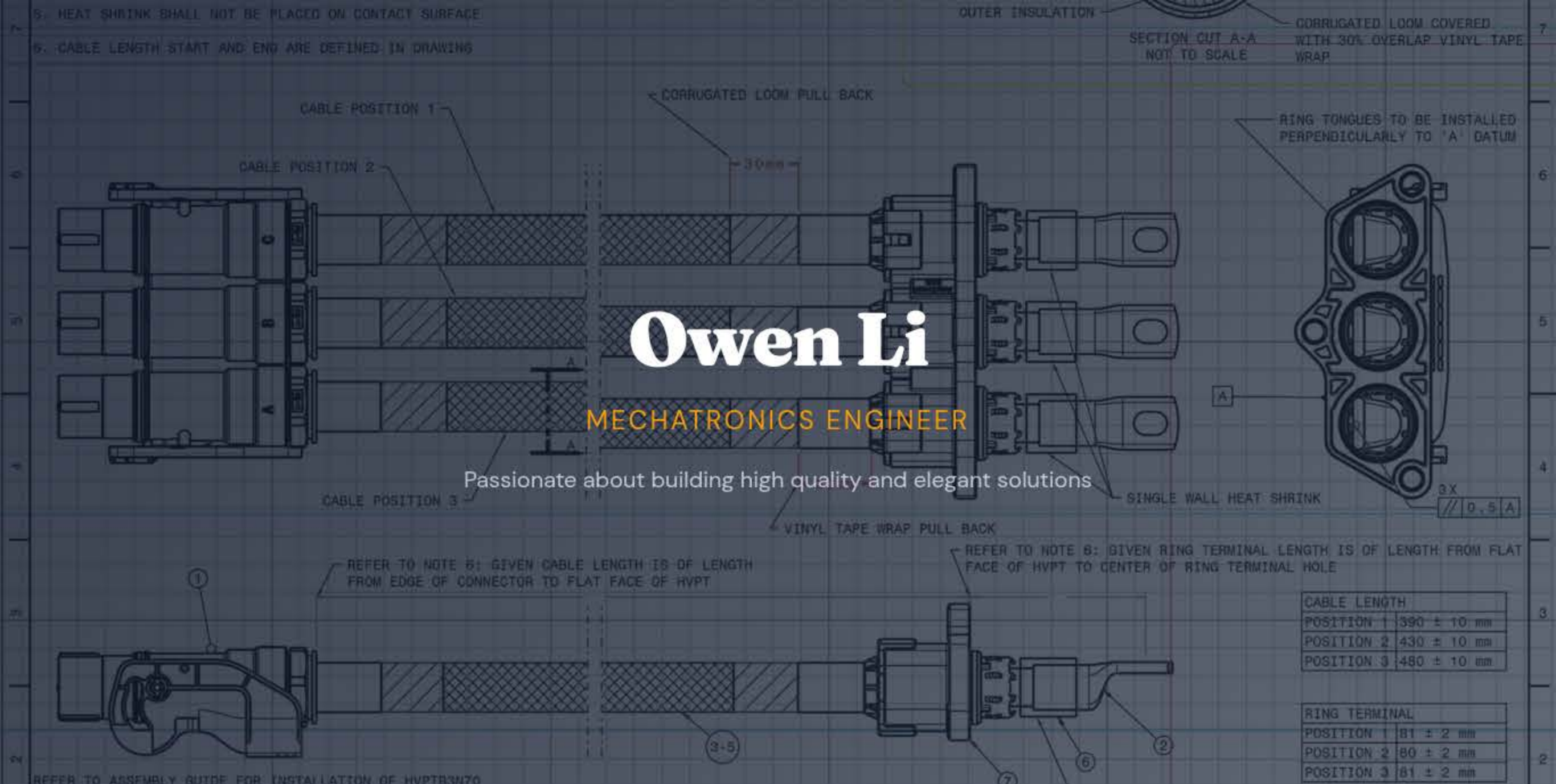
Software: Python, C++

Education

The University of Waterloo

Bachelor of Applied Science in Mechatronics Engineering

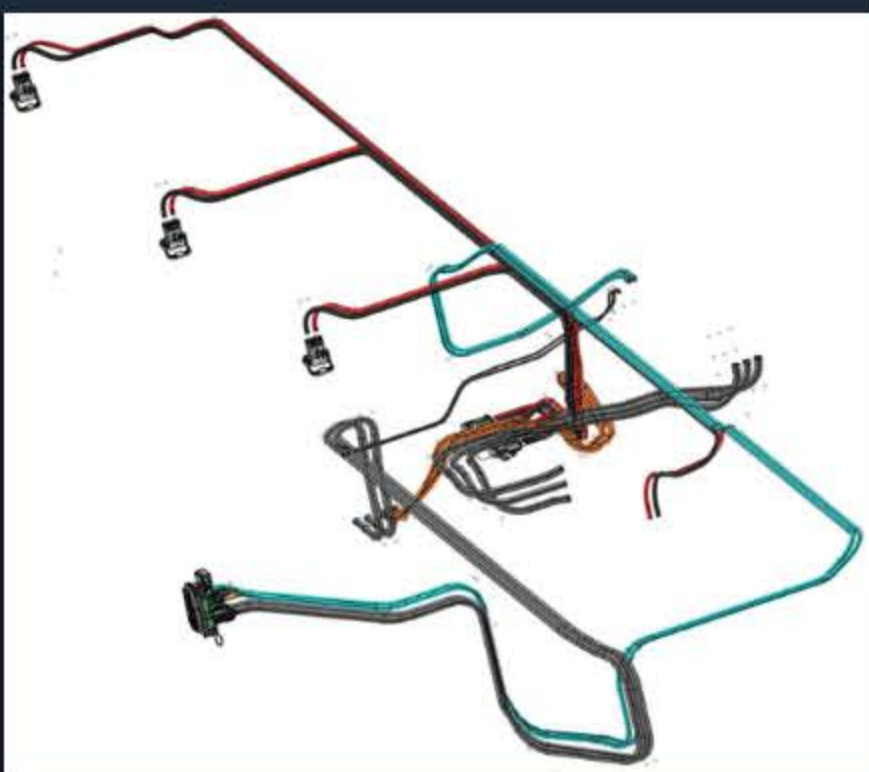
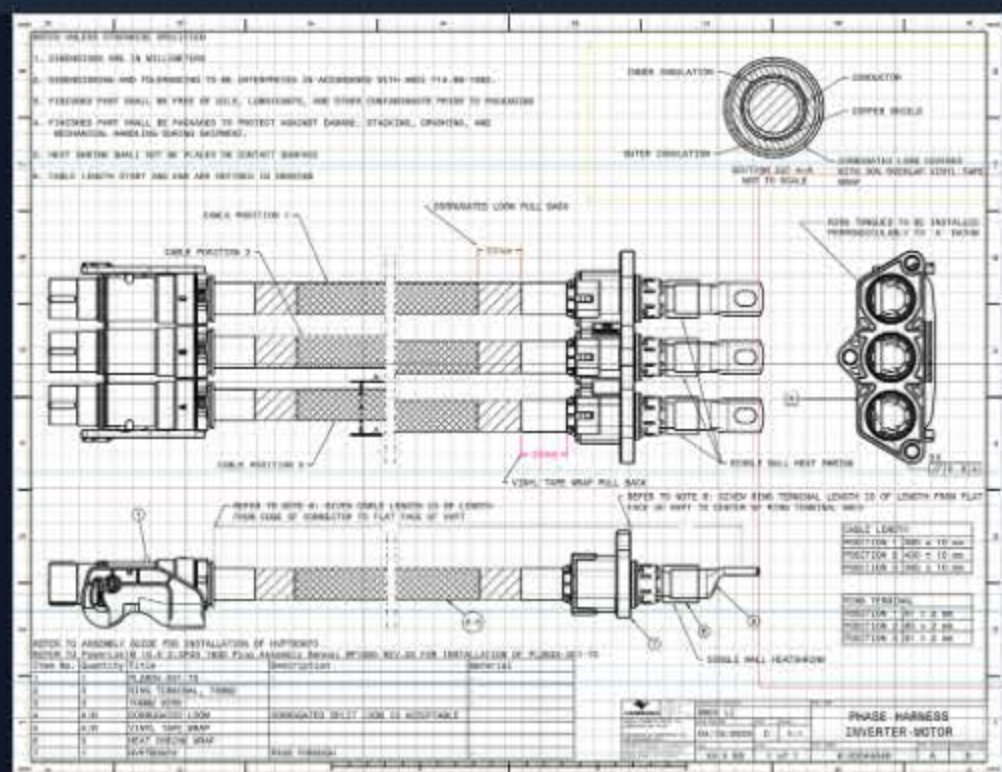
[Graduating December 2026]



About Me

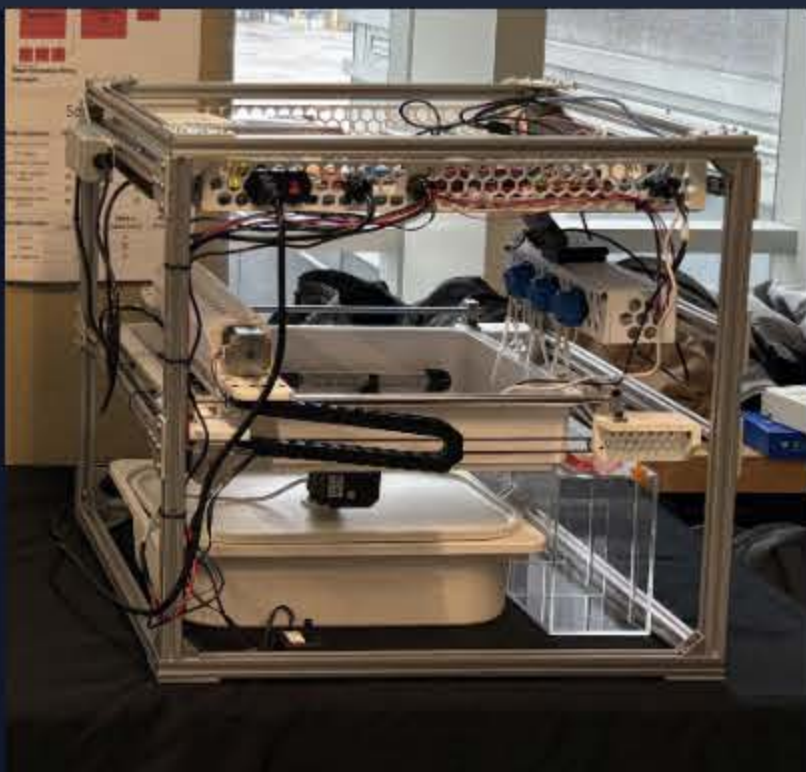
Hello! I'm Owen, a final year Mechatronics Engineering student from The University of Waterloo. I have two years of experience from five co-ops and a design team, with work that spans electric vehicle batteries, wire harness design, and industrial battery testing.

Past Work



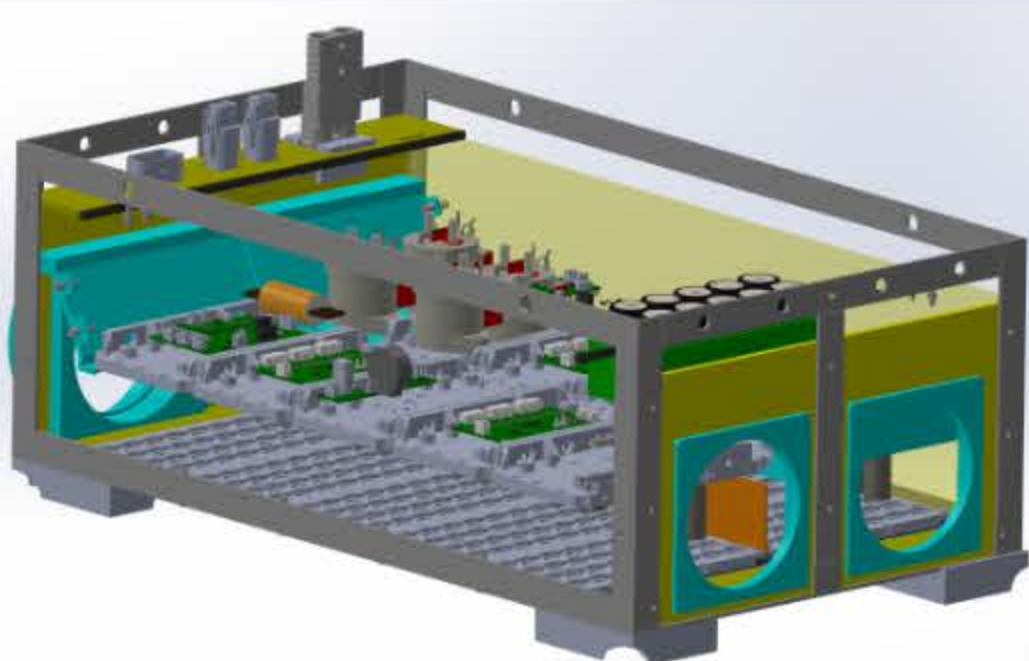
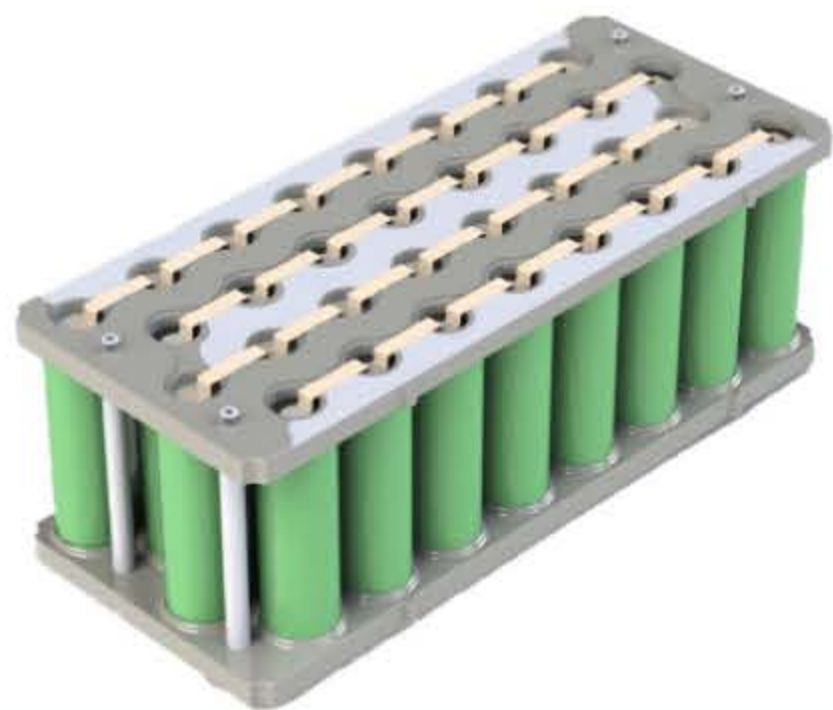
Wire Harness Drawings

- Modeled full-vehicle high-voltage wire harness layouts in CATIA across 3 vehicle wheelbase configurations.
- Authored engineering drawings for wire harnesses and communicated specifications directly with suppliers to obtain quotes and samples.



Automated Wolffia Greenhouse

- Designed end-to-end electrical architecture spanning 5V logic, 12V actuation, 24V pump, and AC mains subsystems for an automated growth and harvest machine for the aquatic plant Wolffia.
- Specified and selected relays, MOSFETs, and buck converters to meet power, switching, and thermal requirements within budget.
- Awarded Mechatronics Class of 2026 Sustainability Prize



Electric Vehicle Battery Design

- Led spec, design, manufacture, and testing of a 36S8P, 5.2 kWh, 131V nominal lithium-ion pack
- Architected modular design of nine 4S8P modules in series with forced-air cooling, packaging of BMS PCBs, relays, fuse, and auxiliary battery into a full electrical assembly
- Performed internal-resistance and thermal characterization of cells and modules; manufactured pack via spot welding to nickel strip with water-jet-cut nickel/copper composite busbars and machined copper busbars.